The spread of the Goosander (*Mergus merganser)* population in north-western Italy

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Abstract – After the first breeding in Italy in 1996 the Goosander has rapidly expanded its breeding range in northern Italy: we analyze the progressive spreading of Goosander in the lakes and rivers, in a wide belt ranging from Torino province to the west, to Garda lake on the east. The study period covers the nesting season from 2010 to 2017. Every water body was assigned to a local coordinator that oversaw the monitoring operation during the breeding season, through a standardized census in the first week of June. The monitoring operation were carried out from the coastline of the water bodies or with the help of wildlife ranger boats belonging to provincial police. We found 282 broods (or families) in 8 different water bodies and the bulk of the nesting pairs was concentrated in the largest lakes, Verbano, Lario and Benaco; we started from 12 broods in 2010 to reach 51 in 2017. The number of chicks per brood ranged between 1 and 19, with a mean value of 6,9: we take the value of 14 chicks as a threshold value of intra-specific parasitism as suggested by Eriksson & Niitylä (1985). From 2011 onward the number of families was almost stable on Verbano lake, revealed a non-significant decrease in Lario lake, whereas on Benaco lake the nesting pairs revealed a statistically significant steady increase. The data collected confirm a successful and ongoing colonization process, harbinger of further spread.

INTRODUCTION

The Goosander breeds in a wide area of central and northern Europe, ranging from latitude 50° to 70° N. Below such limit, in the seventy's, the species was confined in the Swiss lakes and, in Germany, in the Baviera region (Cramp & Simmons, 1977,) were is known a biogeographically isolated population (Hefni-Gautschi et al., 2008). Scattered couples are found in France, Slovenia, Czeck Republic, Ukraina and, in South Europe, in Albania and Greece (Hagemeijer & Blair, 1997). The Goosander nests in holes (Cramp & Simmons, 1977), utilizing both natural and artificial cavities, located preferably in cliffs and walls, without disregarding buildings, and holes derived from Black Woodpecker's nests in central and north Europe; alternatively may use next-boxes. Scarcity or low number of nesting sites may trigger intra-specific competition between females hence favouring the use of the same nest: it's known (Geroudet, 1985) a nest with 39 eggs. This species has synchronous hatching and it's established that broods may be gathered where numerous (Cramp & Simmons, 1977): a threshold value (14 chicks) has been proposed for intra-specific parasitism (Eriksson & Niitylä, 1985). In Italy, at the beginning of XX° century, the Goosander was considered a rare vagrant (Giglioli 1889, Arrigoni degli Oddi 1929), appearing only during migration and in winter: the few data were limited to Garda lake, along the Po river and in the Venetian lagoon (Arrigoni degli Oddi, 1929). From the seventy's onwards, till the start of the 21th century, the Alpine population of Baviera and Switzerland began to grow (BirdLife 2004) and the positive trend spread southward: the first couple was found nesting in Italy in 1996, in the small artificial Corlo's lake (Zenatello *et. al*., 1997), followed by a new one in the western side of Maggiore lake in 1998 (Bordignon, 1999). The two nesting locations are far apart (around 200 km), with many potentially favourable areas, lakes and rivers, in the middle. In 2002 the species was found further east in northern Italy, in the region Friuli Venezia Giulia, along the Isonzo river that runs along the Slovenian border (Felcher & Utmar, 2004) and, from 2000 to 2008, in small lakes and rivers (Piave and Brenta) of the Alpine sector of Veneto and Friuli Venezia Giulia (Zenatello *et al*., 2009). Very probably these scattered pairs have different origins in the European breeding area, underlining the active and easily observable dispersion dynamism of the individuals, matched with an increasing numbers of wintering flocks in the pre-Alpine lakes of northern Italy. For example, the wintering numbers on Verbano and Lario lakes, passed from a none/few individuals in 2002 to a three digits figure in a 15 years period (Vigorita *et al*., 2002; Longoni & Fasola, 2016). The breeding of the species , from the start of the new century, revealed a positive trend, colonizing others lakes and river in northern Italy: in 2003 the first broods were observed in the eastern (Gagliardi *et al*., 2007) and northern (Volet & Burkhardt, 2003) part of Lake Maggiore, followed by the first one in 2004 on Iseo Lake (Bordignon *et al*., 2009) joined in 2005 by the very first on Como Lake (Viganò *et al*., 2006). Subsequently, further east, nesting pairs were found in 2010 (Gargioni & Piotti, 2013) on the western coast of Garda Lake (the largest lake in Italy), followed respectively in 2014 and 2015, in the northern coast (Volcan G., pers.comm.) and eastern one (Sighele 2018). Meanwhile in 2013 was discovered the first broods in the Sarca river, 5 km north of Garda Lake (Volcan G., pers.comm.). Evidence of new breeding pair arrived in 2015 from Toce river (Tosi A., pers. comm.), tributary on the western side of Maggiore Lake, while in 2017 was discovered a brood in the small Garlate lake (Viganò E., pers.comm.), very close to the southern side of Como lake. Last but not least, once again during 2017, arrived the confirmed breeding of a pair on Orta Lake (Tosi A., Bordignon L., in press), the westernmost one in northern Italy. In this water body a family was already observed in September 1998, but the young were already flying (Bordignon, 2004). The progressive colonization of the species prompted the creation of a voluntary group in 2010, the "Goosander Group", formed by several ornithologist and birdwatcher in northern Italy, to census the breeding population.

MATERIAL AND METHODS

Study area

The research was conducted, from 2010 to 2017, in a wide area (approximately of 400x 200 kilometres) in north-Western Italy, comprising 16 lakes and 6 main rivers (Table 1) of the Piedmont, Lombardy, and partly Trentino Alto Adige and Veneto regions concerning Garda lake, stretching from the pre-Alpine mountains to the Padana Plain, north of river Po (Figure 1). The westernmost tip correspond to Avigliana lake (n. 1, Lat: 45.053 N, Long: 7.390E) is in the Piedmont region, whereas the easternmost is located at the northern tip of Garda lake (n. 15, Lat: 45.867, Long: 10.876) in Trentino Alto Adige. The northernmost part is located at Mezzola lake (n. 7, Lat: 46.212 N, Long: 9.445 E), just north of Como lake; the southern one at the Mantova lakes (n. 16, Lat: 45.146 N, Long: 10.809E)

lakes				rivers		
		mean altitude	surface (km^2)			lenght (km)
1	Avigliana	354	1,5	**17**	Dora Baltea	168
2	Candia	226	1,52	**18**	Sesia	140
3	Viverone	230	5,8	**19**	Ticino	110
4	Orta	290	18,2	**20**	Adda	164
5	Maggiore (*Verbano*)	193,8	212	**21**	Oglio	95
6	Lugano (*Ceresio*)	270	48,9	**22**	Mincio	75
7	Mezzola	200	5,9			
8	Como (*Lario*)	199	145			
9	Montorfano	397	0,46			
10	Alserio	260	1,23			
11	Pusiano	257	5,2			
12	Annone	224	5,71			
13	Garlate	198	4,64			
14	Iseo (*Sebino*)	185	65,3			
15	Garda (*Benaco*)	65	370			
16	Mantova lakes	18	6,21			

Table 1: lakes and rivers that were monitored in the period 2010-2017, respectively ordered from west to east. * In Maggiore Lake were monitored the main tributaries too

The lakes ranges from very small bordered with reed beds and woods, with low depth, to the biggest ones in Italy: Orta, Maggiore, Lugano, Como, Iseo and Garda (this one with the largest surface) are characterized mainly by mountain rocky coasts, dotted with built up areas overlooking the lake borders, with only a small amount of reed beds/woods, generally at the southern tip.

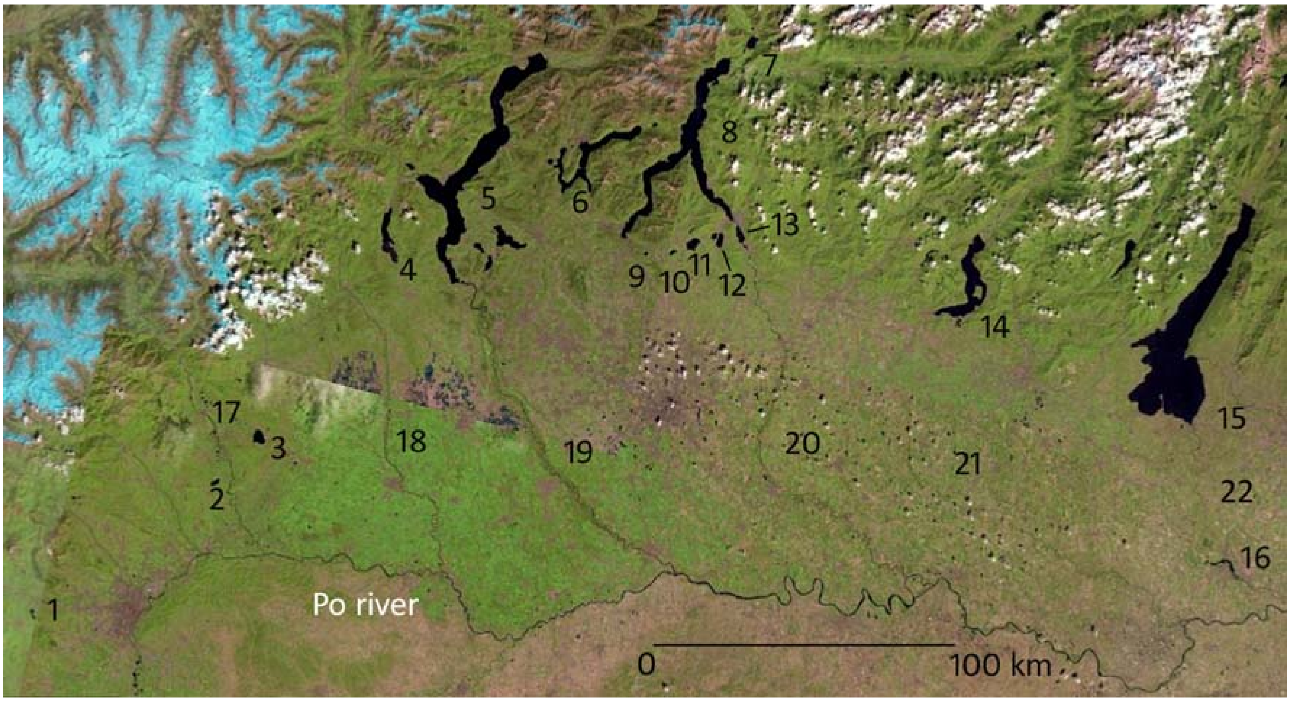

Figure 1: map of the study area in northern Italy, based on a Landsat 8 composed images dated 30th August 2017, with the 16 lakes and 6 river numbered progressively from west to east. Accessed on 19 February 2018 at www.Landsatlook.usgs.gov

Methods

Aim of this research was the monitoring of the breeding population, from 2010 till 2017, to delineate the trend in different areas with the census of females with chicks (brood size). Our census protocol required a local coordinator for every water body: for the largest ones (Maggiore, Como and Garda), the western and eastern sides were further subdivided among a minimum of two or more local coordinator, to cover the entire length of the suitable coasts; in the case of Verbano lake were was a further subdivision because the northern part belongs to Switzerland involving local manpower too. In all the water bodies the broods were actively searched every year from May to July: this time frame encompasses the main period of the appearance of the family parties, in agreement with the census work already carried out in the nearby Switzerland (Keller & Gremoud, 2003). Further, every year, was conducted a contemporary census among the 22 areas in the first week of June, to obtain as far as possible a correct figure of the number of broods, that was utilized to compare the evolution of the progression of the colonization in the various water bodies. The only exception to this protocol regards the Verbano lake, on the Lombardy' s side from 2015 to 2017, and the Swiss part from 2010 to 2017, where observed families where gathered in the period May-July. To avoid the chance of double counts in different days, for assessing the number of broods we considered only the first observation of a family in the same place, in view of the fact that generally the number of chicks, in the growing phase, tends progressively to decrease. In this way we believe that the number of observed families may be underestimated, anyway preferring a little underestimation than double (or more) repeated counts. The shores of lakes and rivers were actively searched from vintage points with binoculars and telescopes by monitoring group formed by a minimum of two people; some years, in the biggest lakes, we also have had support from local gamekeepers that made available a patrolling boat to scan the coast. During the census the position of every female with chicks was mapped: we consider a brood or family (female with chicks) as those one formed by chicks of the same age, assessing different families if the observed age was clearly detectable. Every year, at the end of the monitoring session, the counts were transmitted to the general coordinator (Lucio Bordignon) to store the data. Graphics and statistical analysis were carried out with R software (The R Foundation; www.r-project.org).

RESULTS

In 8 years we found 282 broods in 8 water bodies, with a steady increase (Figure 2) statistically significant (linear model R = 0,8074, p= 0.0024**), starting with 12 families in 2010 and arriving to a maximum of 51 in 2017 (Table3). Only the four major lakes (Maggiore, Como, Iseo and Garda) regularly occupied; in 2013 families were observed for the first time in two rivers (Ticino and Adda, respectively n. 19 and 20) and, from 2015 till 2017, in the river Toce too, that belongs to the water catchment of lake Maggiore. In 2016 was observed a territorial pair along the Sesia river (n. 18) and again in 2017 two pairs in 2017, but no evidence of breeding was discovered in both years. In 2017 families were discovered in the Orta (n. 4) and Garlate (n.13) lakes: in the first site already in 2015 were observed pair without proof of breeding. The Garlate lake is a small lake, at the southern tip of Como lake, with heavily urbanized shores.

year	2010	2011	2012	2013	2014	2015	2016	2017
n. families	12	28	28	31	49	41	42	51

Table 3: counted number of families (female with chicks) for year

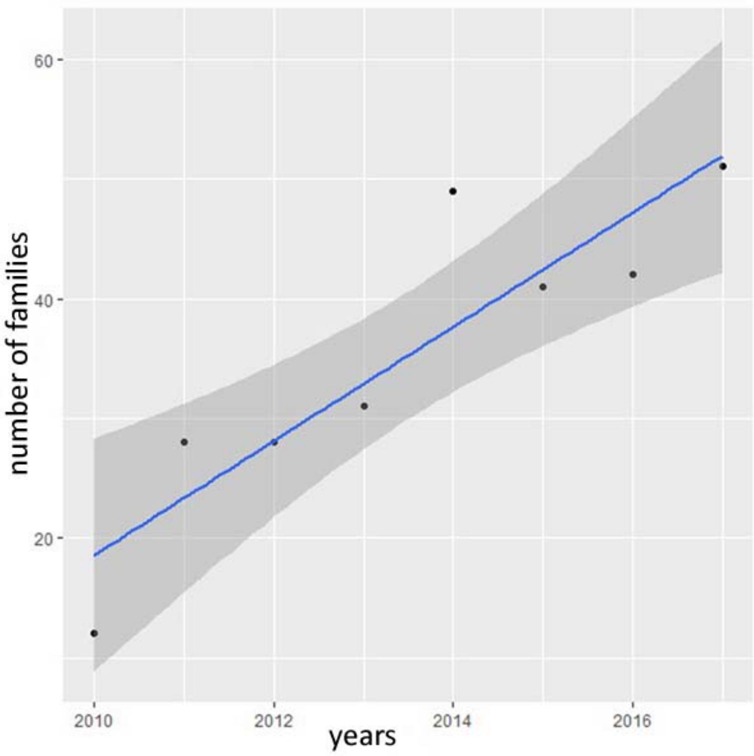

Figure 2: regression line for the total number of families per year

From 2010 to 2014 the bulk of population (84.3%) was concentrated in two lakes, Maggiore (n.5) e Como (n.8), with other sites representing the 15.7% (Table 4); the median value of the number of broods doesn't differs between the two lakes (Mann-Whitney test; W = 19, p= 0.1867, two-sided).The Iseo lake (n. 14) has always hosted a low number of families during the study period, with a maximum of 3 families in 2016, whereas the Garda lake (n. 15) showed a positive trend (Figure 3), starting with the first observed brood in 2010 , arriving at 16 families in 2017 (Mann-Kendall test; tau = 0.945, p= 0.0022**). The other three sites (n. 5, 8, 14) doesn't reveal a statistically significant trend, with fluctuating number of broods, more sharper on Como lake.

	Verbano **(n. 5)**	Lario **(n. 8)**	Sebino **(n. 14)**	Benaco **(n. 15)**
2010	6	4	1	1
2011	14	12	1	1
2012	9	15	2	2
2013	10	14	2	2
2014	22	21	2	4
2015	11	19	2	7
2016	13	15	3	10
2017	14	15	1	16
total	99	115	14	43
mean	12,4	14,4	1,6	5
ds	4,75	1,79	0,52	1,8
median	12	15	2	3
tau (M-K test)	0,473	0,491	0,229	0,945
p value	0,1346	0,1265	0,5623	0,0022 **

Table 4: total number of broods counted in the four major lakes, mean (ds) and median. Tau value for Mann-Kendall (M-K) test (two sided) for the 8 years trend, with p value

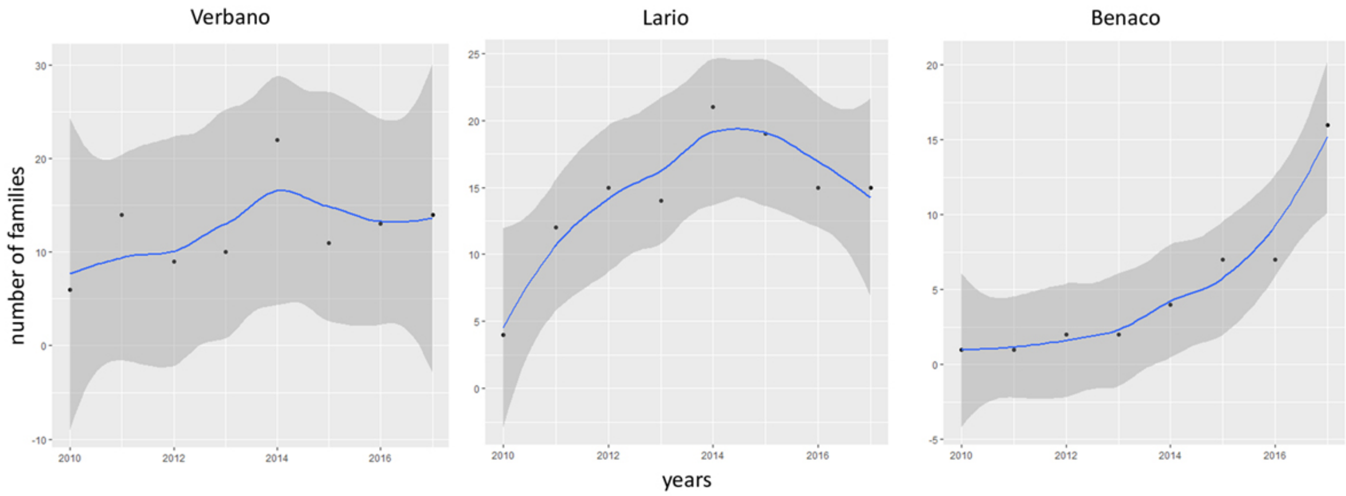

Figure 3: trend-lines with smooth regression (± 95% CL) of the number of families in Verbano, Lario and Benaco lakes

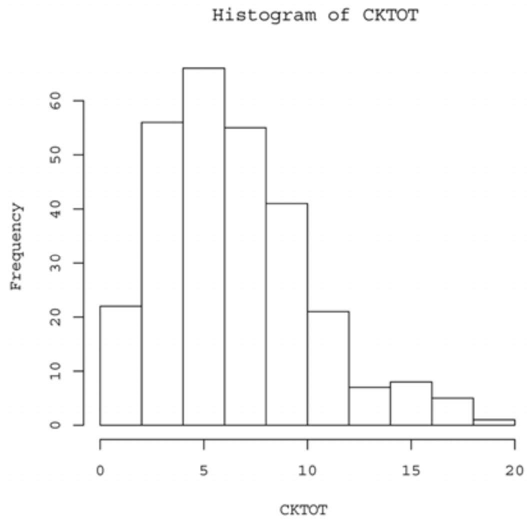

Figure 4: frequency distribution of the number of chicks per broods

The number of chicks per brood (n= 282) ranges from 1 to 19, with a mean value of 6.94 (sd = 3.64) and a median value of 6: the frequency distribution is right-skewed (Figure 4; mode= 5). Broods with a number of chicks up to 9 are 221 (78,4%), whereas greater than or equal to 10 are 61 (21,6%); broods with a number of chicks equal or higher than the threshold value for intra-specific parasitism (≥ 14) are 15 (5.31%). The median value of chicks per broods decreases in first three years, stabilizing from 2013 onwards (Figure5), however there isn't a significant difference in the median values per brood per year (Kruskal-Wallis test H=11,972, d.f.=7, p=0,1015)

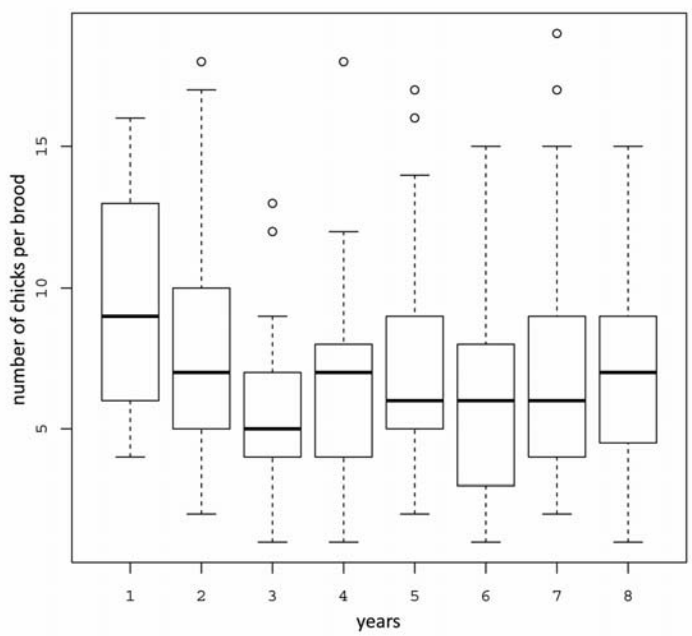

Figure 5: box plot of the number of chicks per brood per year (1=2010, 8=2017)

DISCUSSION

Concerning the relatively recent colonization of the Italian water bodies, Gustin *et al.* (2016) could not provide a Favourable Reference Value for the Goosander in Italy: following the European guidelines for assessing the conservation status and habitat, they calculated a Minimum Viable Population of 300 pairs, target value to reach a stable breeding population; in our view this value is absolutely attainable following the population trend of the last few years. At the European level (EU 27) the Goosander belongs to the lowest category of threat "Least Concern" but, the short-term population trend is reported (BirdLife 2015) as "decreasing", mainly for the decline of the Finland population (-31/45%) in the period 2001-2012; Finland and Sweden represent the northern stronghold of this species. Reviewing the distribution of the species in north-western Italy, we update the known geographic range that, at the end of the 2017 breeding season (www.ornitho.it; accessed on 28 February 2018), regards only the northern Italy, from west to east, reaching the Slovenian border. The update involves the Italian population estimate too that, from the last published reference (Nardelli *et al*., 2015) was assessed at 22-29 pairs for the period 2008-2012. Only for our study area, considering the successful pairs of 2017, we have a minimum of 51 females with chicks: in the north-eastern Italy, a little lower figure of successful females may be obtained (www.ornitho.it; accessed on 28 February) for the same year, but we can assess a minimum of 80-150 nesting pairs for the whole country, quadrupling the previous published figure. From 2015 onward the breeding population in the western lakes (Verbano and Lario) appears to be stable: in the first lake the Goosander is becoming increasingly frequent in the southern part of the water body, that presents different characteristics than the northern sector, where the rocky coast is predominant. In the southern part, partly urbanized, the species is more frequent too (Saporetti, 2018) in the areas surrounding reeds of the Special Protection Area "Canneti del Lago Maggiore", a marshland-woodland site. In 2017 a brood (Ravasi, pers.comm.) was observed close to the Angera harbor, a town bordering the marshland area "Palude Bruschera". North of Garda lake, in Trentino Alto Adige, out of our study area, in the Valsugana Valley along the Brenta river (east of Trento), different broods (with a total of 45 chicks) were observed from 2015 to 2017. Outside our survey but in the same years, others families were found in Lombardy on Brembo river (Bergamo province, San Gervasio municipality): in 2015 (Santinelli pers.com.) observed, on 15 June, a female with 9 chicks; close to the previous place, on Adda river (Canonica d'Adda municipality), another brood was found in 2017, on 20 June, with a female followed by 24 young (Redaelli, pers.com.). Is very likely that others pairs may have occupied nesting places scattered along rivers that are more difficult to monitor than the lakes, in which the use of the boat may greatly improve the observation of the broods: in any case we expect that soon new places will be included in the Italian breeding range.

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